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Declaration

This report is submitted to the University of Warwick in support for IB9KC: Financial Econometrics module. It has been composed by every member of this group and has not been submitted in any previous application for any degree. **No form of Artificial Intelligence** has been used in producing any parts of this report, which includes the python programming as well. Several python packages has been used in this project and the python syntax and ideas are taken from examples mentioned in their documentation. All the techniques and functions used in both the report and the python program have been implemented after thoroughly studying the lecture notes, seminar and assignment resources and the python packages documentation available on the internet.

All the resources, project data, and Jupyter notebook files can be found on the GitHub repository.

Abstract

This report discusses how well a small number of statistical and theoretical motivated factors can explain the common variations and cross-sectional pricing of the 25 Fama-French size and book-to-market portfolios.

The first part applies Principal Component Analysis (PCA) to the portfolio excess returns. The results show that the three components capture nearly 93% of total variation, with the first component overwhelmingly dominant and closely aligned with the market factor, the second with a size-related dimension, and third displaying a weaker and less stable value type pattern. An out-of-sample construction confirms that the first two components are robust across subsamples, while the third is sensitive to sample splitting, suggesting its economic interpretation should be treated with caution.

The second part runs Fama-MacBeth regressions to test whether these statistical factors, alongside the Fama-French and macroeconomic factor models, actually price the cross-section of returns. The results reveal a clear disconnect between explaining covariance and explaining average returns: although Principal Component Factor 1 (PC1) dominates total variation, it shows no significant cross-sectional premium, while the weaker Principal Component Factor 3 (PC3) linked to value appears to be priced. The out-of-sample PCA model produces the smallest and least significant pricing intercept, suggesting it leaves behind the least common pricing error (no arbitrage) among the four specifications, while the in-sample PCA model delivers the strongest explanatory power in terms of R^2. Although the macroeconomic model produced a comparatively small pricing intercept, this result should not be interpreted as strong performance. The goodness-of-fit of the macroeconomic model suggests that the factors are unable to fully capture the variation in portfolio returns. Moreover, the factors are nearly uncorrelated with the dominant return-based components and only the term spread carries a marginally significance.

The third part reviews Hou *et al.* [1]'s q-factor model, highlighting both its theoretical grounding in q-theory of investment and its empirical ability to digest a wide range of anomalies. We show how the model's investment and profitability factors are motivated by firms' optimal investment conditions, giving the framework a stronger structural basis than earlier factor models built primarily on empirical regularities. We then consider whether this stronger anomaly-digesting performance should be interpreted as evidence in favour of the Efficient Market Hypothesis (EMH). Although the q-factor model reduces abnormal alpha more effectively than earlier benchmark models, this conclusion must be qualified by the joint-hypothesis problem and by the possibility that some apparent risk compensation may instead reflect systematic biases or behavioural mispricing. Lastly, we compare Fama & French [2]'s empirically motivated profitability-factor choice with Hou *et al.* [1]'s theory-driven construction, to show how each paper's preferred profitability proxy reflects its broader methodological objective.

List of abbreviations

CP	Cash Profitability
EMH	Efficient Market Hypothesis
FM	Fama-MacBeth
HML	High Minus Low (book-to-market factor)
OP	Operating Profitability
PC1	Principal Component Factor 1
PC2	Principal Component Factor 2
PC3	Principal Component Factor 3
PCA	Principal Component Analysis
ROE	Return on Equity
SMB	Small Minus Big (size factor)
VIF	Variance Inflation Factor

1 Forming some Factors

1.1 Introduction

This chapter examines, the common variation in size and BE/ME portfolio excess return summarised by a small number of latent factors using PCA[3]. PCA is a standard multivariate dimension-reduction technique that transforms a set of correlated variables into orthogonal components ordered by the amount of variance they explain. In empirical asset pricing, the method essentially captures the dominant sources of return co-movement directly from the data, providing a useful statistical insight to theory-motivated models such as Fama-French, [4]. Principal factors are constructed using two approaches: an in-sample procedure based on the full covariance matrix, and an out-of-sample procedure designed to assess the stability of the extracted factors across subsamples. Their relationships with the Fama–French and macroeconomic factors are then examined in order to assess both economic interpretation and stability.

1.2 In-sample principal components

To construct the in-sample principal components, the sample covariance matrix of excess returns is formed and this sample covariance matrix is used to obtain the eigenvectors and eigenvalues. The eigenvectors associated with the 3 largest eigenvalues of the covariance matrix are the principal component factors. The associated eigenvalues measure how much of the total variance is explained by each component, and the eigenvectors provide the portfolio weights used to construct the corresponding factor portfolios [3].

The covariance matrix is predominantly positive. This indicates that there is substantial co-movement across the 25 portfolio excess returns, suggesting the presence of at least one strong common component affecting most portfolios. The largest covariance and variance values appear among the small-stock portfolios, with the small low-BE/ME portfolio recording the highest variance (figure 1.1). The magnitude of covariances declines as we move from small to large size portfolios, with differences across BE/ME groupings also present, though less pronounced. These features suggest that portfolio returns are driven by common factors rather than moving independently, providing a clear motivation for applying PCA as a dimension-reduction tool [4].

The first three principal components explain approximately 84.6%, 4.8% and 3.5% of the total variation, respectively, indicating that the structure of covariance is dominated by a single common factor. The corresponding in-sample factor time series, formed by applying the estimated eigenvector weights to the full sample of excess returns, supports this conclusion. The other two principal components therefore appear to capture weaker residual dimensions of cross-sectional co-movement, potentially related to firm characteristics such as size or book-to-market. Overall, a small number of latent statistical factors can summarise the dominant common variation across the 25 portfolio returns.

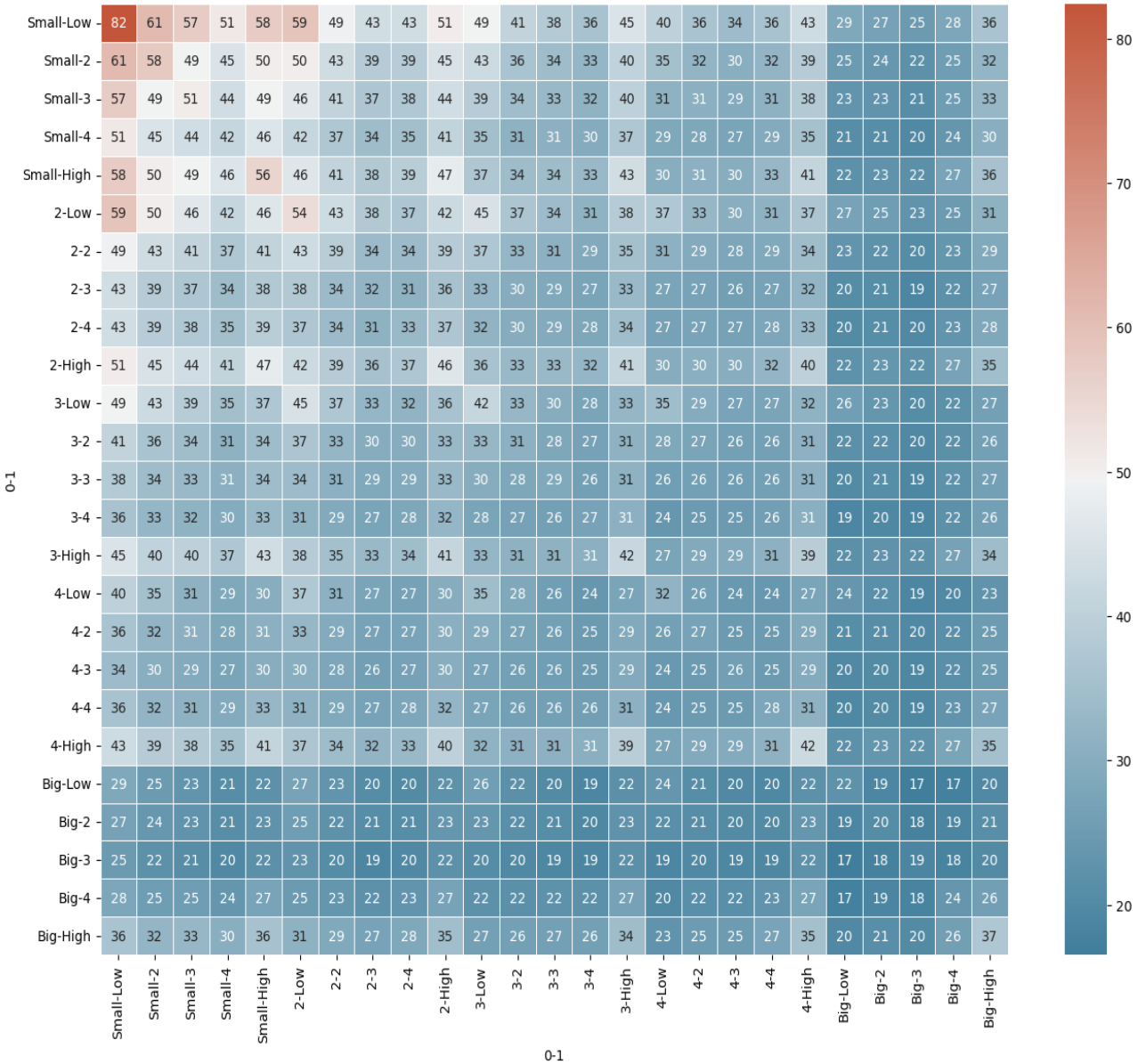

Figure 1.1: *Monthly Returns Correlation*

Table 1.1: *Eigenvectors and Eigenvalues of the Even Portfolios*

	Book-to-market equity (BE/ME) quintiles									
Size	Low	2	3	4	High	Low	2	3	4	High
quintile			w_1 [PC1]					w_2 [PC2]		
Small	0.2877	0.2571	0.2286	0.2164	0.2217	-0.4713	-0.3107	-0.1712	-0.1193	-0.0646
2	0.2515	0.2159	0.1971	0.1943	0.2183	-0.2350	-0.0791	0.0106	0.0424	0.0819
3	0.2218	0.1946	0.1827	0.1720	0.2062	-0.1584	0.0459	0.1088	0.1565	0.1596
4	0.1935	0.1806	0.1706	0.1742	0.1973	-0.0869	0.1152	0.1761	0.1695	0.2659
Big	0.1475	0.1456	0.1338	0.1427	0.1635	0.0724	0.1991	0.2259	0.2935	0.3758
			w_3 [PC3]					Eigenvalue		
Small	-0.0608	-0.1203	-0.1523	-0.2254	-0.3383	675.1073	48.2190	24.1468	10.7977	7.3646
2	0.2169	0.0514	-0.0519	-0.1266	-0.2591	5.5949	4.3324	4.1030	3.5890	2.9992
3	0.3238	0.1345	0.0109	-0.0329	-0.2103	2.8015	2.6046	2.3360	2.1906	2.0158
4	0.4107	0.1790	0.0657	-0.0258	-0.1967	1.7696	1.5506	1.5225	1.4105	1.3796
Big	0.3929	0.2293	0.1661	0.0318	-0.0979	1.3250	1.1782	1.1119	0.9521	0.8923

1.3 Out-of-sample principal components

The stability of the PCA factors can be examined by constructing out-of-sample principal components. The excess-return data are split into odd- and even-months - the eigenvectors estimated from the odd-month covariance matrix are applied to the even month-returns, while eigenvectors estimated from the even month covariance matrix are applied to the odd-month returns. These two sets of factor returns are then combined to form a full out-of-sample factor series. Since each month's factor is constructed using weights estimated from a different subsample rather than from the same observations used to generate that return, the procedure guards against look-ahead bias and provides a more rigorous test of whether the extracted factors reflect genuine, stable sources of systematic risk rather than sample-specific noise [4].

Table 1.2: *Eigenvectors and Eigenvalues of the Odd Portfolios*

	Book-to-market equity (BE/ME) quintiles									
Size	Low	2	3	4	High	Low	2	3	4	High
quintile			w_1 [PC1]					w_2 [PC2]		
Small	0.2830	0.2426	0.2437	0.2179	0.2608	-0.4321	-0.2234	-0.2567	-0.1754	-0.3364
2	0.2358	0.2118	0.1928	0.1996	0.2371	-0.0931	-0.0281	-0.0069	0.0082	-0.0763
3	0.2002	0.1837	0.1808	0.1749	0.2194	0.0589	0.1298	0.1252	0.1097	0.0367
4	0.1655	0.1669	0.1635	0.1739	0.2182	0.2024	0.2010	0.2317	0.1754	0.1357
Big	0.1277	0.1281	0.1205	0.1488	0.1947	0.2932	0.2958	0.2640	0.2276	0.0980
			w_3 [PC3]					Eigenvalue		
Small	-0.2724	-0.1147	0.0561	0.0879	0.2644	921.0038	44.8771	36.8636	12.0285	8.2680
2	-0.3244	-0.1270	-0.0389	0.0684	0.1944	6.2032	4.9892	4.2621	3.9062	3.3812
3	-0.3935	-0.1263	0.0202	0.1008	0.2908	2.5882	2.4366	2.2676	2.1905	2.0537
4	-0.3367	-0.0631	0.0111	0.1341	0.2689	1.9399	1.6425	1.5278	1.4493	1.3100
Big	-0.2058	-0.0940	0.0000	0.2038	0.3164	1.2106	1.1546	1.0711	1.0556	0.9469

The eigenvectors from the table 1.1 and table 1.2 provide further economic interpretation of the principal components. In both odd- and even-months, the first eigenvector assigns positive weights across all 25 portfolios, which is consistent with a broad market-wide factor affecting all portfolios in the same direction. The second eigenvector displays a clearer contrast across the size dimension, with smaller portfolios generally receiving opposite-signed weights to larger portfolios, suggesting Principal Component Factor 2 (PC2) likely captures size-related factor. By contrast, the third eigenvector is less stable across the subsamples. In the odd-month sample it shows a more evident BE/ME pattern, where we can see a clear gradient across this dimension with negative weights on low-BE/ME portfolios and positive weights on high-BE/ME portfolios indicating that PC3 captures a value versus growth component. While, for even-month sample, the same pattern is no longer observed. The loadings are uniformly negative for the small-stock portfolios but mostly positive for the medium- and large-size portfolios, suggesting the component reflects a mizture of size-related structure. However, the pattern across BE/ME categories are not consistent enough to interpret the component as a pure-value factor. The eigenvalues reinforce this interpretation, since the first eigenvalue is overwhelmingly larger than the others in both subsamples, whereas the second and third components are much weaker and therefore more susceptible to sampling variation.

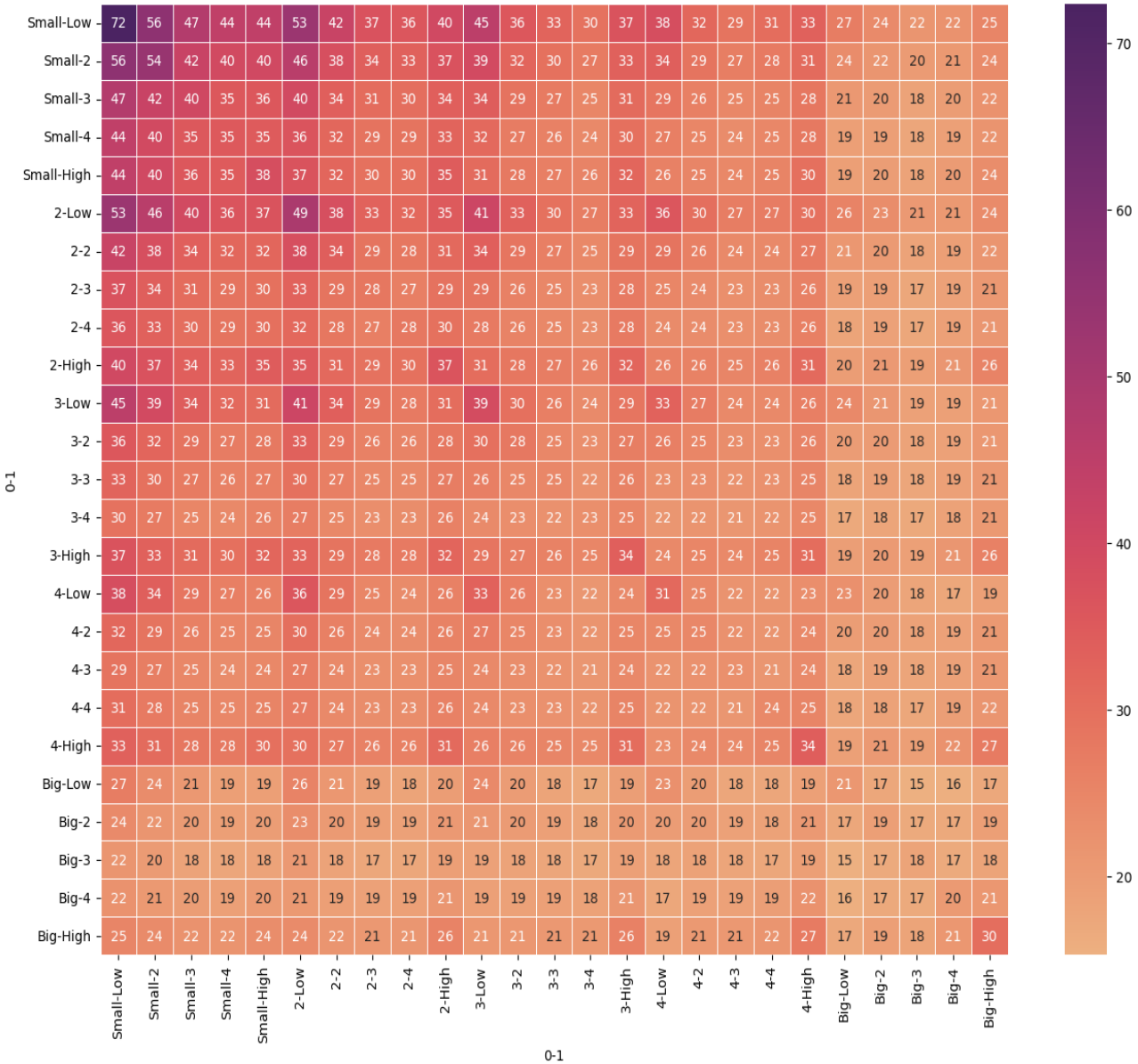

Figure 1.2: *Even Monthly Returns Correlation*

In both, the odd- and even-month subsamples, the largest diagonal and off-diagonal entries are concentrated among the small-stock portfolio, indicating that these portfolios are both more volatile and strongly linked in absolute terms. The big-stock portfolios, by contrast display lower covariance values overall, suggesting lower volatility and weaker joint variation. Similar to the previous matrix (figure 1.1), differences in BE/ME categories are also present, but less pronounced. The similarity of these covariance patterns across subsamples indicates that the main structure of return variation is stable over time.

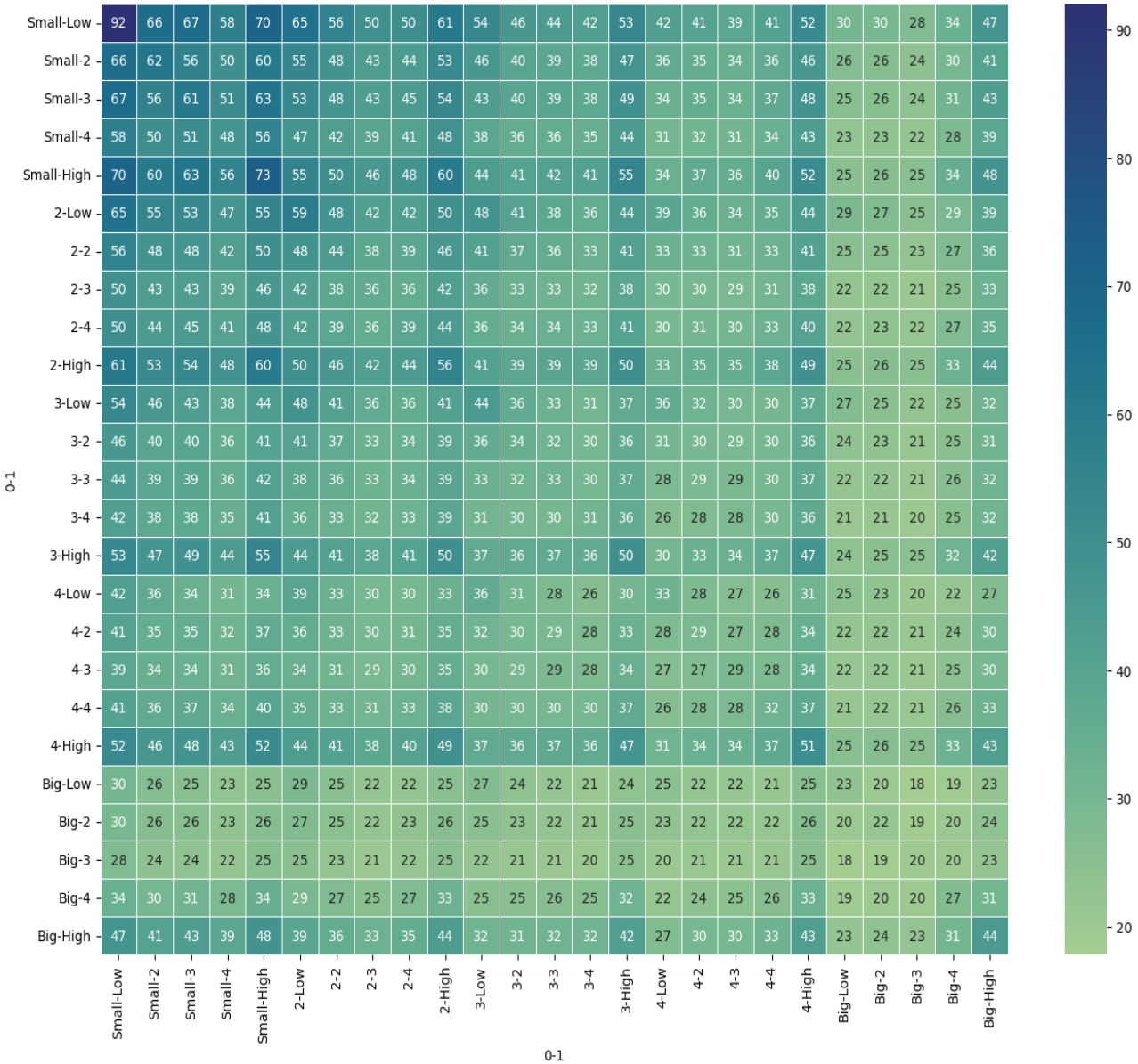

Figure 1.3: *Odd Monthly Returns Correlation*

1.4 Correlation matrix of Fama-French, macroeconomic, and PCA factors

The figure 1.4 reports the correlation matrix between the Fama-French factors, the macroeconomic factors and the PCA factors. Observe that in-sample and out-of-sample PC1 have a nearly perfect correlation and in-sample PC2 has a very high correlation with out-of-sample PC2. This suggests that the first two principal components are highly stable across the two constructions. This is not surprising since the first two components explain the largest shares of total variation and should therefore be the least sensitive to sample selection.

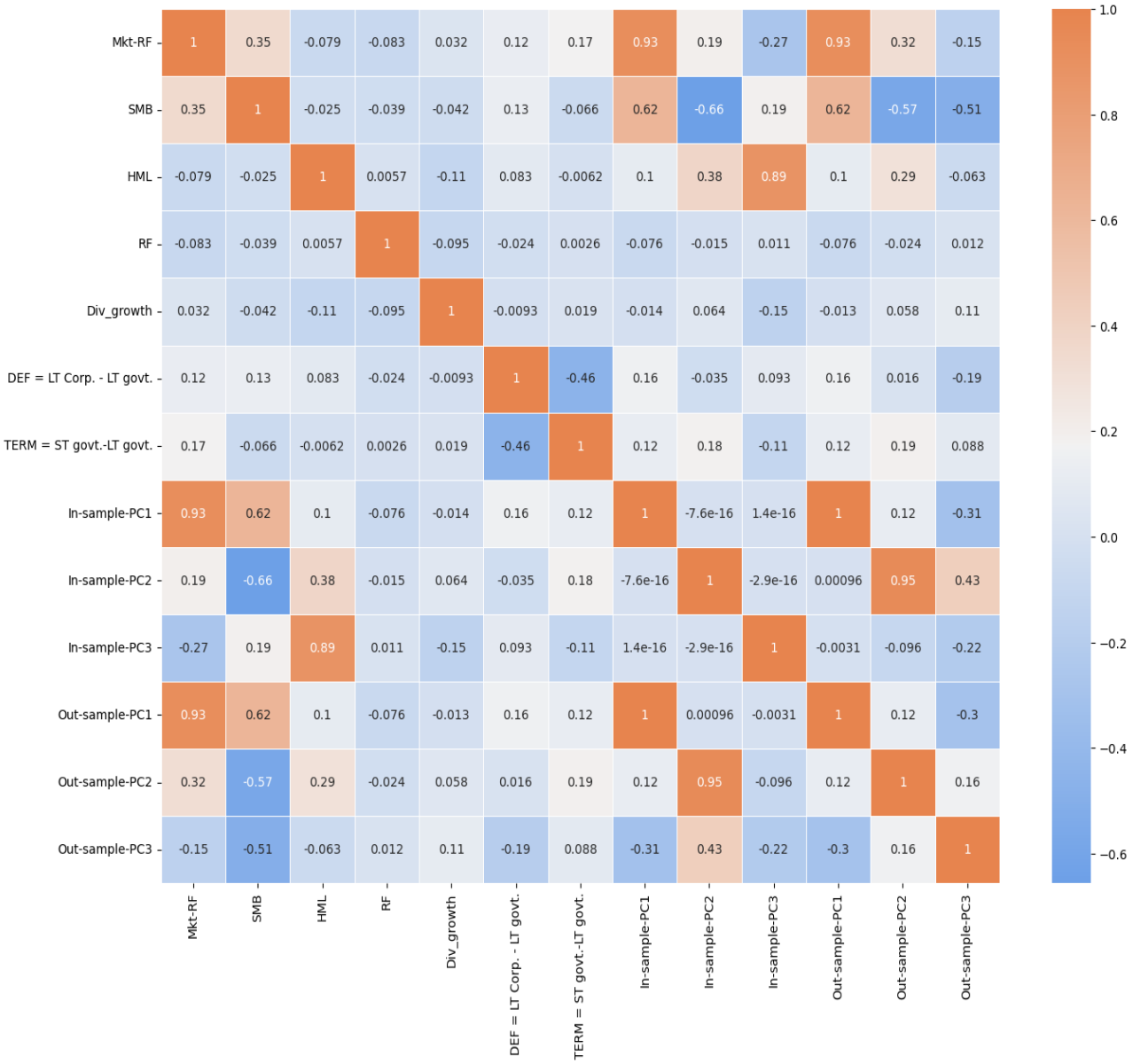

Figure 1.4: *Correlation Matrix including all Factors*

By contrast, the correlation between in-sample PC3 and out-of-sample PC3 is weak and negative indicating the third component is less stable. This is also not very surprising, since higher order principal components explain only a small fraction of total variation and therefore more sensitive to variation in samples[3]. The figure 1.4 also demonstrates extremely small cross-correlations between some of the principal components, i.e., in-sample PC1 is essentially uncorrelated with in-sample PC2 (-7.6 $\times 10^{-16}$) and in-sample PC3 (1.4 $\times 10^{-16}$). The in-sample PC2 is likewise uncorrelated with the in-sample PC3 (-2.9 $\times 10^{-16}$). These near-zero values are exactly what we expect from PCA, since the principal components are constructed to be orthogonal within sample. Out of sample, however, the exact orthogonality no longer holds, the out-of-sample PC3 is more strongly correlated with in-sample PC2 (0.43) than with in-sample PC3 (-0.22), suggesting that weaker higher-order components are more sensitive to sample splitting and their interpretation may shift across subsamples. This is because higher-order principal components explain only a small fraction of total variation and are therefore more vulnerable to estimation error and sampling noise. When the sample is split, these weaker components are less precisely identified, so their loadings and economic interpretation can change more easily than those of the dominant first two components.

Turning to economic interpretation of these factors, the market factor Mkt-RF is very strongly correlated with both in-sample PC1 and out-of-sample PC1, with correlations of around 0.93 in each case. This strongly suggests that the first principal component captures broad market-wide movements in returns. This interpretation is economically intuitive since, if most of the 25 portfolios co-move because of aggregate market shocks, the first statistical factor extracted from the covariance matrix should resemble the market factor. The result also aligns closely with the Fama–French framework, in which the market factor plays the central role in explaining common variation in shock returns[5].

The second principal component appears to be closely related to the size factor. Both in-sample PC2 and out-of-sample PC2 are fairly strongly negatively correlated with Small Minus Big (size factor) (SMB) (-0.66 and -0.57 respectively). The negative sign is not itself economically important, since principal components are only identified up to sign; reversing the sign of an eigenvector yields the same component in some sense. What matters is the strength of the relationship, which indicates that the second principal component is capturing a size-related dimension of return variation. This is an interesting result, because it shows that a purely data-driven PCA procedure is recovering a factor closely aligned with the Fama-French size factor[5].

In-sample PC3 is strongly positively correlated with High Minus Low (book-to-market factor) (HML), suggesting it may reflect a value-related factor. However, this interpretation does not survive out of sample. Out-of-sample PC3 has only a weak relationship with HML, but a moderately strong negative correlation with SMB, suggesting that it is more closely related to a residual size-type effect than to the value effect captured by in-sample PC3. This helps explain the weak correspondence between in-sample and out-of-sample versions of PC3 and the moderately strong relationship between out-of-sample PC3 and PC2. The contrast with the robustness of the market and size dimensions suggests that the value effect, while identified by Fama and French (1993)[5] as a systematic factor, is less reliably recovered through purely statistical methods.

The macroeconomic factors, by contrast shows only weak relationships with both the Fama-French factors and principal components. While Dividend growth is essentially uncorrelated with the dominant return-based factors, with correlations of only 0.032, and -0.014 with Mkt - RF and in-sample PC1, respectively. This suggests that it does not line up closely with the main dimensions of portfolio co-movement identified here. Similarly, default spread and term spread show only small correlations with the main market-related component, implying that the dominant source of common variation in the 25 portfolio returns is not primarily driven by the selected macroeconomic variables. The same conclusion is seen for the higher-order principal components. The most notable relationship within the macroeconomic block itself is the moderately negative correlation between default spread and term spread (-0.46), suggesting that these variables contain related but opposing information about macro-financial conditions, even though neither maps closely onto the dominant statistical factors extracted from the portfolio returns.

Overall, the evidence indicates that the dominant common variation in the 25 portfolio returns is much more closely aligned with market, size, and to a lesser extent value-type return factors than with the selected macroeconomic factors. This is consistent with the Fama-French finding that equity returns are primarily driven by a small number of return-based common factors [5], and with the broader result in Connor and Korajczyk (1986)[4] that statistical factor methods recover economically meaningful dimensions of return co-movement directly from the covariance structure of returns.

2 Asset Pricing Tests

2.1 Introduction

This chapter examines the performance of four models - Fama-French, the macroeconomic factors model, in-sample PCA, and out-of-sample PCA - in explaining the cross-section of portfolio returns. The Fama-MacBeth procedure is a two-step method used to study whether expected returns are related to systematic risk exposures. In the first step, each portfolio's exposure to the relevant factors is estimated from a time-series regression of excess returns on the chosen factor(s). In the multifactor case, the cross-sectional pricing relation can be written as:

$$E[R_i] - r_f = \gamma_0 + \boldsymbol{\gamma}'\boldsymbol{\beta}_i + \eta_i$$

where β_i denotes the vector of factors tested for portfolio i.

The first-stage regression yields the portfolio's sensitivity to each factor. In the second step, the cross-section of returns is related to the estimated betas. In the full Fama-MacBeth approach, this cross-sectional regression is run repeatedly over time and the resulting slope coefficients are averaged to obtain the estimated prices of risk. A simplified version of the procedure is also implemented in which average portfolio returns are regressed directly on the estimated betas. In this sample, both versions produce very similar results because they rely on the same first-stage betas and yield numerically equivalent risk-premium estimates, with differences arising only in the standard errors.

2.2 Empirical Analysis

This section presents the empirical results of the Fama-MacBeth regressions for the four models under consideration. The models are compared using three criteria: the magnitude and significance of the pricing intercept, the statistical significance of the estimated risk premia, and the average R^2 of the first-stage time-series regression. Together, these measures assess both cross-sectional pricing performance and overall explanatory power.

2.2.1 Pricing intercepts and cross-sectional pricing performance

The pricing intercept γ_0 provides a direct measure of how well each model prices the cross section of returns. If a factor model fully captures the cross-sectioinal variation in expected excess returns, γ_0 should be close to zero and statistically insignificant. A large or statistically significant intercept suggests that the model leaves behind a common pricing error.

Under this criterion, the Fama-French model produces a positive and highly significant intercept, indicating that the factor loadings on the market, size, and value factors do not fully explain the cross section of returns on the 25 portfolios.

The in-sample PCA model performs better but its intercept is still statistically significant. This means that the in-sample PCA factors, despite being very successful at summarising common return variation in Part I, still leave a non-trivial common pricing component unexplained. This is

not a contradiction. PCA is constructed to maximize explained covariance, not to maximise the fit of average returns and ensure that the extracted factors earn priced premia in the cross-section.

The macroeconomic model generates a smaller intercept that is statistically insignificant suggesting a weaker common pricing error. However, this result should be interpreted with caution, since the intercept only indicates whether the model leaves a large common pricing error or not. It does not imply that the individual macroeconomic factors are important in pricing the portfolios.

The out-of sample PCA performs the best under the γ_0 criterion. It has a statistically insignificant intercept with the smallest absolute value. This suggests that among the four specifications, it leaves behind the smallest common pricing error.

The intercept results suggest that the out-of-sample PCA model performs best under the γ_0 criterion, followed by the macroeconomic model, then the in-sample PCA model, with the Fama-French model performing worst on this particular measure. Also taking into consideration the relatively large standard deviations of the intercept estimates, it can be inferred that pricing errors vary considerably over time, such that none of the models is able to provide a consistently stable pricing relation.

Table 2.1: *Constants of Different Models - Descriptive Statistics*

	Fama-French Factors	**Macroeconomic Factors**	**In-sample Model**	**Out-sample Model**
Count	870.00	870.00	870.00	870.00
Mean	1.7018	-0.4649	0.7810	0.3920
Std	8.3502	8.9043	7.0674	6.7824
Min	-35.8850	-45.1903	-33.1986	-30.4168
25%	-2.7989	-5.3207	-3.3874	-3.2077
50%	1.5374	-0.1792	0.6014	0.1012
75%	6.3362	4.9780	5.1658	4.7721
Max	39.5375	36.1810	32.3035	30.5072

2.2.2 Estimated factor premia

The vector of risk premia $\boldsymbol{\gamma}$ measures the price of risk associated with each factor, and therefore indicates whether a factor is priced in the cross-section of returns.

In the Fama-French model, the market factor and HML are statistically significant, while SMB is not. The significance of HML makes sense, given that the test portfolios are sorted by BE/ME and Part I linked in-sample PC3 to a value-type dimension. A surprising result here appears to be the negative market premium. Within this set of 25 equity portfolios, market betas display relatively limited cross-sectional dispersion compared with the size and value loadings. As a result, the second-pass regression may attribute a greater share of cross-sectional return differences to HML and SMB-related structure than to the market factor itself. Additionally, in Part I, PC1 was clearly market-like, yet in the in-sample PCA model, PC1 is not statistically significant, while Fama-French market factor is. This shows that the extracted market-like component is not identical to the Fama-French market factor in cross-sectional pricing terms. The two may capture similar common co-movement over time, while differing in the way their beta variation maps into average returns once the model is estimated.

The macroeconomic model provides only weak evidence of priced risk. Among the three macro factors, only the term-spread factor is statistically significant. This is interesting to note in conjunction with the relatively small and insignificant intercept observed earlier. Although the

model does not leave a large common pricing error, it still does a poor job of producing individually important priced factors. This matches chapter 1 results, where macroeconomic factors were only weakly related to the dominant PC factors.

For the in-sample PCA model, only PC3 is statistically significant. This result illustrates the difference between explaining covariance and pricing average returns. In chapter 1, PC1 explained the overwhelming majority of total variation and PC2 was also relatively stable, yet neither earns a significant cross-sectional premium here. By contrast, PC3 explains much less variation but appears priced. The natural interpretation is that PC3 captures a dimension along which the portfolios differ more sharply in average returns. In chapter 1, we have linked the third in-sample principal component to a value-type pattern, its significance is consistent with the idea that value-related differences matter more for the cross section of expected returns than the dominant market-like movement captured by PC1.

In the out-of-sample PCA model, PC2 and PC3 are significant. The significance of PC2 and PC3 is economically plausible since our analysis in chapter 1 suggested that PC2 was closely related to size. This makes the comparison with the Fama-French model especially interesting: the PCA-based size-related factors appear significant but SMB is not. This could indicate that PCA factors may be extracting extra size-related cross-sectional patterns that are present in the data but not captured clearly by the standard SMB construction. It is also important to recall that out-of-sample PC3 is the least stable. Therefore, it is plausible that instead of being a robust new priced factor, PC3 is capturing some residual cross-sectional structure that just happens to be priced in this sample.

Table 2.2: *Fama-French Model Results*

	Mean	**Std Dev**	**Std Error**	***t*-stat**
γ_0	1.7018	8.3502	0.2831	6.0113
γ_1	-1.0205	9.2431	0.3134	-3.2564
γ_2	0.1122	3.1058	0.1053	1.0654
γ_3	0.4477	2.9814	0.1011	4.4290

Table 2.3: *Macroeconomic Model Results*

	Mean	**Std Dev**	**Std Error**	***t*-stat**
γ_0	-0.4649	8.9043	0.3019	-1.5400
γ_1	-0.0007	0.0122	0.0004	-1.7159
γ_2	0.0023	0.0725	0.0025	0.9462
γ_3	0.0163	0.1867	0.0063	2.5724

Table 2.4: *In-Sample Model Results*

	Mean	**Std Dev**	**Std Error**	***t*-stat**
γ_0	0.7810	7.0674	0.2396	3.2597
γ_1	0.3939	44.7073	1.5157	0.2599
γ_2	0.2619	8.8484	0.3000	0.8730
γ_3	0.8439	6.3966	0.2169	3.8914

Table 2.5: *Out-of-Sample Model Results*

	Mean	**Std Dev**	**Std Error**	***t*-stat**
γ_0	0.3920	6.7824	0.2299	1.7049
γ_1	2.1596	43.5522	1.4766	1.4626
γ_2	1.6192	12.0766	0.4094	3.9547
γ_3	-3.2906	24.0294	0.8147	-4.0392

2.2.3 Interpreting the average R^2

The coefficient of determination, R^2, measures how well the model explains variation in portfolio returns. This is a time-series fit criterion, not a pricing criterion. Hence, a model with a high R^2 can still leave a non-negligible pricing intercept.

The in-sample PCA has the highest average R^2, which is exactly what one would expect from chapter 1. The factors are constructed directly from the full-sample covariance matrix to capture as much common variation in returns as possible. This is why strong R^2 performance of the in-sample PCA specification should be recognised with care as it might indicate over-fitting as opposed to a good model.

The out-of-sample PCA has a slightly lower R^2 compared to in-sample PCA but still performs strongly overall. This again matches chapter 1, the first two out-of-sample principal components were highly similar to the in-sample versions, but the third component was less stable. So, a modest reduction in fit is expected. Although the model preserves the main factor structure, it does not exploit the full sample as aggressively as the in-sample PCA construction.

The strong R^2 performance of the Fama-French model is consistent with chapter 1. There PC1, PC2, and the in-sample PC3 aligned broadly with the market, size and value. So the Fama-French model is capturing much of the same dominant structure. The relatively small standard deviation of R^2 for FF and in-sample PCA additionally suggests that their explanatory power is consistently high across portfolios.

The macroeconomic model has a very low average R^2 value of $\approx 6.7\%$, suggesting that macro variables are very weak predictors of portfolio returns in this sample. This is highly consistent with chapter 1, where the macro factors had weak correlations with both the PCA factors and the Fama-French factors. Also, that fact that macroeconomic factors are slow moving, which capture time series variation in the aggregate economy where portfolios returns are cross sectional risk premia driven. Hence, they are not closely linked to the dominant covariance structure of returns, implying that they explain very little of the time-series variation in the 25 portfolios.

Under the R^2 criterion, the in-sample model showcases the best performance, followed closely by the Fama-French and then the out-of-sample model. The macroeconomic model has the lowest mean R^2 value and provides little explanation for this sample.

2.2.4 Concluding Remarks

A strong fit does not necessarily translate into good pricing. This is particularly evident in the in-sample PCA model: although it fits the returns well, its significant intercept indicates that it does not fully price the cross-section. A strong model should ideally perform well on both assessed criteria, R^2 and γ_0. Here, no model dominates both criteria, but the out-of-sample PCA model provides the most balanced performance, combining relatively low pricing errors with strong explanatory power.

Additionally, capturing the covariance structure of returns and explaining the cross-section of expected returns prove to be two distinct, albeit connected, objectives. The main dimensions identified in chapter 1 are therefore useful for understanding the data, but only some of them translate into priced risk in chapter 2.

Table 2.6: *Descriptive Statistics of Model R^2 Across 25 Iterations*

	Fama-French Factors	**Macroeconomic Factors**	**In-sample Model**	**Out-sample Model**
Count	25.00	25.00	25.00	25.00
Mean	0.9116	0.0672	0.9252	0.8900
Std	0.0306	0.0136	0.0305	0.0540
Min	0.8262	0.0400	0.8434	0.7308
25%	0.9007	0.0578	0.9200	0.8761
50%	0.9120	0.0657	0.9367	0.9163
75%	0.9372	0.0779	0.9439	0.9276
Max	0.9545	0.0922	0.9565	0.9399

Table 2.7: *Fama-French Three-Factor Regressions*

Regressions of excess stock returns (in percent) on the Fama-French three factors.

$$\mathrm{R}(t) - \mathrm{RF}(t) = a + b_{\mathrm{MKT}}[\mathrm{RM}(t) - \mathrm{RF}(t)] + b_{\mathrm{SMB}}\,\mathrm{SMB}(t) + b_{\mathrm{HML}}\,\mathrm{HML}(t) + e(t)$$

	Book-to-market equity (*BE/ME*) quintiles									
Size quintile	Low	2	3	4	High	Low	2	3	4	High
			b_{MKT}					b_{SMB}		
Small	1.1776	1.0661	1.0189	0.9488	1.0246	1.6029	1.3315	1.2076	1.0753	1.2528
2	1.1465	1.0228	0.9508	0.9733	1.0983	1.0958	0.8985	0.7588	0.7292	0.8857
3	1.0968	1.0259	0.9941	0.9606	1.1124	0.7581	0.5134	0.4419	0.3856	0.5868
4	1.0641	1.0440	1.0205	1.0229	1.1853	0.3551	0.2195	0.1600	0.2356	0.3245
Big	1.0248	1.0009	0.9346	1.0095	1.1385	−0.2032	−0.2215	−0.2355	−0.1833	−0.0142
			b_{HML}					R^2		
Small	0.0587	0.2047	0.4758	0.5375	0.9277	0.8262	0.8904	0.9039	0.9397	0.9275
2	−0.1920	0.1377	0.3176	0.5234	0.8367	0.9219	0.9372	0.9389	0.9459	0.9545
3	−0.3703	0.1093	0.3676	0.5195	0.8815	0.9435	0.9134	0.9102	0.9106	0.9214
4	−0.3824	0.1627	0.3335	0.5419	0.9039	0.9329	0.9091	0.8939	0.9007	0.8872
Big	−0.2879	0.0599	0.2320	0.6429	0.9817	0.9428	0.9120	0.8625	0.9061	0.8567

The market betas b_{MKT} are all positive and tightly clustered around 1. This supports the inference that the negative risk premia observed for the market factor - (γ_1) - isn't really a contradiction of theory. More likely, the Fama-French size and value factors capture the main cross-sectional differences in the portfolio returns, with market beta adding little independent explanatory power once SMB and HML are included. Additionally, there is a negative relationship between size and return: b_{SMB} is higher for small portfolios and generally declines as size increases. b_{HML} risees with BE/ME and is strongest for high book-to-market portfolios. The Fama-French model thus displays a reasonably strong performance, with good R^2 values.

Table 2.8: *Macroeconomic Factor Regressions*

Regressions of excess stock returns (in percent) on macroeconomic factors.

$$\mathrm{R}(t) - \mathrm{RF}(t) = a + b_{\mathrm{DIV}}\,\mathrm{Div_growth}(t) + b_{\mathrm{DEF}}\,\mathrm{DEF}(t) + b_{\mathrm{TERM}}\,\mathrm{TERM}(t) + e(t)$$

	Book-to-market equity (*BE/ME*) quintiles									
Size quintile	Low	2	3	4	High	Low	2	3	4	High
			b_{DIV}					b_{DEF}		
Small	−225.8700	−137.4408	−190.0531	−210.3060	−180.9726	171.7683	157.6465	161.6265	147.8709	185.4945
2	82.4274	4.8983	−29.4511	−63.4577	−78.3402	155.9505	137.0203	132.9437	143.3125	158.5306
3	95.7474	51.8816	17.5131	5.8320	−96.2487	128.9234	121.4645	128.5424	131.2528	151.1828
4	132.8298	53.0640	29.9633	−20.0475	−16.5169	108.3347	121.6842	113.8185	124.8615	152.6092
Big	99.2632	49.0950	20.1800	−46.9418	−107.5556	88.3227	93.7948	100.4994	120.2685	135.1968
			b_{TERM}					R^2		
Small	57.7866	53.4011	57.4656	52.2411	54.8279	0.0400	0.0460	0.0578	0.0606	0.0640
2	58.2222	57.4558	59.5287	63.2576	62.0063	0.0496	0.0561	0.0678	0.0754	0.0613
3	58.4446	60.2036	62.6429	62.8189	68.5660	0.0509	0.0649	0.0768	0.0835	0.0691
4	59.0420	64.6641	65.9011	67.3616	73.2305	0.0586	0.0811	0.0821	0.0831	0.0723
Big	55.0435	58.0606	60.2107	60.6322	55.2912	0.0657	0.0779	0.0922	0.0848	0.0576

The macroeconomic factor loadings do not exhibit a clear monotonic pattern across the size and BE/ME portfolios, and the R^2 values are uniformly low, indicating that the macro model explains very little of the cross-sectional variation in excess returns in this sample.

Table 2.9: *In-Sample Principal Component Regressions*

Regressions of excess stock returns (in percent) on in-sample principal components.

$$\mathrm{R}(t) - \mathrm{RF}(t) = a + b_{\mathrm{PC1}}\,\mathrm{PC1}(t) + b_{\mathrm{PC2}}\,\mathrm{PC2}(t) + b_{\mathrm{PC3}}\,\mathrm{PC3}(t) + e(t)$$

	Book-to-market equity (*BE/ME*) quintiles									
Size	Low	2	3	4	High	Low	2	3	4	High
quintile			b_{PC1}					b_{PC2}		
Small	0.2854	0.2491	0.2377	0.2176	0.2448	−0.4941	−0.2917	−0.1971	−0.1322	−0.1405
2	0.2426	0.2136	0.1948	0.1976	0.2295	−0.2225	−0.0779	−0.0085	0.0335	0.0390
3	0.2093	0.1883	0.1816	0.1738	0.2140	−0.1255	0.0594	0.1158	0.1475	0.1485
4	0.1772	0.1727	0.1666	0.1741	0.2096	−0.0108	0.1416	0.1962	0.1905	0.2450
Big	0.1360	0.1355	0.1260	0.1462	0.1816	0.1381	0.2219	0.2396	0.2962	0.3024
			b_{PC3}					R^2		
Small	−0.0080	0.0248	0.1609	0.1739	0.3811	0.9220	0.9198	0.9367	0.9517	0.9565
2	−0.2349	−0.0826	−0.0002	0.0818	0.2321	0.9439	0.9419	0.9393	0.9398	0.9480
3	−0.3557	−0.1579	−0.0293	0.0413	0.2303	0.9546	0.9393	0.9362	0.9282	0.9388
4	−0.4050	−0.1613	−0.0835	0.0319	0.1948	0.9534	0.9449	0.9320	0.9214	0.9200
Big	−0.3438	−0.2181	−0.1349	0.0436	0.2156	0.8722	0.9014	0.8434	0.8895	0.8553

The in-sample principal components capture the common variation in the 25 size–BE/ME portfolios very well, as expected, with R^2 values generally above 0.87. PC1 has positive loadings across all portfolios and declines with size, suggesting a broad common factor with stronger exposure among small firms. PC2 seems to be representing a size-based factor with loadings on portfolios exhibiting an increasing trend when going from small to big. PC3 is most strongly associated with the BE/ME dimension, with high loadings on small value portfolios and negative loadings on many large or low-BE/ME portfolios.

Table 2.10: *Out-of-Sample Principal Component Regressions*

Regressions of excess stock returns (in percent) on out-of-sample principal components.

$$\mathrm{R}(t) - \mathrm{RF}(t) = a + b_{\mathrm{PC1}}\,\mathrm{PC1}(t) + b_{\mathrm{PC2}}\,\mathrm{PC2}(t) + b_{\mathrm{PC3}}\,\mathrm{PC3}(t) + e(t)$$

	Book-to-market equity (*BE/ME*) quintiles									
Size	Low	2	3	4	High	Low	2	3	4	High
quintile			b_{PC1}					b_{PC2}		
Small	0.2882	0.2502	0.2343	0.2158	0.2379	−0.4760	−0.2825	−0.1830	−0.1340	−0.1374
2	0.2464	0.2153	0.1965	0.1973	0.2272	−0.1990	−0.0708	−0.0171	0.0237	0.0220
3	0.2155	0.1908	0.1820	0.1723	0.2091	−0.1073	0.0621	0.1118	0.1436	0.1406
4	0.1812	0.1733	0.1678	0.1722	0.2072	0.0202	0.1519	0.1887	0.1876	0.2210
Big	0.1387	0.1387	0.1270	0.1429	0.1773	0.1603	0.2192	0.2385	0.2952	0.2856
			b_{PC3}					R^2		
Small	−0.1478	−0.0984	−0.1241	−0.0830	−0.1558	0.9176	0.9200	0.9276	0.9347	0.8877
2	−0.0291	−0.0110	0.0110	−0.0022	−0.0312	0.9003	0.9349	0.9399	0.9322	0.9083
3	0.0396	0.0515	0.0395	0.0246	−0.0256	0.8501	0.9186	0.9360	0.9225	0.8900
4	0.0569	0.0588	0.0812	0.0361	0.0410	0.7912	0.9240	0.9293	0.9163	0.8761
Big	0.0921	0.1219	0.0981	0.0576	0.0391	0.7308	0.8568	0.8309	0.8792	0.7943

The out-of-sample principal components continue to explain a substantial portion of the variation in the 25 size–BE/ME portfolios, with R^2 values generally ranging from around 0.85 to

0.94. The fit is weaker than in the in-sample case, supporting the interpretation that some of the in-sample explanatory power reflects overfitting. PC1 is positive across all portfolios and declines with size, again suggesting a broad common factor with strong exposure among small firms. PC2 appears to broadly resemble a size-related factor. PC3 is less structured but still contributes to explaining residual variation across portfolios, this matches chapter 1 observations.

3 Digesting Anomalies

3.1 Introduction

Hou, Xue, and Zhang's "Digesting Anomalies: An Investment Approach"[1] proposes a 4-factor model founded on the q-theory of investment whereby firms' expected returns should be correlated with the magnitude of their investments and their profitability. Throughout the paper, the investment and profitability factors are both empirically and theoretically rationalized as an improved factor-model relative to both the Fama-French and the Carhart models:

$$E[r^i] - r^f = \beta^i_{\text{MKT}} E[\text{MKT}] + \beta^i_{\text{ME}} E[r_{\text{ME}}] + \beta^i_{I/A} E[r_{I/A}] + \beta^i_{\text{ROE}} E[r_{\text{ROE}}]. \tag{3.1}$$

- $E[r^i] - r^f$: expected excess return of asset i
- $\beta^i_\bullet$: $\bullet$ factor sensitivity of asset i
- **Factor premiums**: MKT (market), r_{ME} (size) , $r_{I/A}$ (investment), r_{ROE} (profitability)

The factors are constructed from a triple $2 \times 3 \times 3$ sort on size, investment to assets, and ROE, with factors returned as the difference between the high and low portfolios within each sort.

Under this framework, the firm's optimal investment condition equates the marginal cost of investment to the discounted marginal benefit of future cash flows. This generates the central q-theory prediction (see eq. 4 in Hou *et al.* [1]) that, holding expected profitability fixed, firms that invest more must face lower expected returns. Conversely, holding investment fixed, firms with higher expected profitability should have higher expected returns.

Intuitively, the investment and profitability rate reveal the discount rate implicit in the valuation of future cash flows. High investment signals relatively cheap capital, since a high required rate of return would depreciate current value too much to justify high investment. Conversely, low investment signals higher required return. For a given level of investment, stronger expected profitability must therefore be associated with higher expected returns. This gives the investment and Return on Equity (ROE) factors a structural role in the proposed factor model. This mechanism echoes the dividend discount intuition formalized by [2], who show that expected profitability and expected investment are both priced in the cross-section even within a present-value framework. When empirically tested against the preexisting Fama-French and Carhart models the proposed "q-factor model" performs better at explaining the 35 statistically significant anomalies found in the broad cross section:

Metric \ Model	q-factor	Carhart	Fama-French
Average absolute unexplained alpha per month	0.20%	0.33%	0.55%
Number of anomaly sets rejected by the GRS test (out of 35)	20	24	28
Number of statistically significant unexplained anomalies (out of 35)	5	19	27

Table 3.1: *Empirical comparison between factor models. [1]*

Before narrowing to the 35 significant anomalies, Hou *et al.* [1] examine nearly 80 anomalies drawn from the literature, finding that approximately half are statistically insignificant in the broad cross section, suggestion that many proposed anomalies are exaggerated or sample specific.

3.1.1 Relation to Efficient Market Hypothesis

Under the EMH, asset prices fully reflect all available information; therefore, persistent abnormal alpha should not exist. The q-factor model is consistent with a rational, efficient market interpretation that the investment and profitability factors should be understood as compensation for systematic risk rather than evidence of mispricing. For instance, firms with high profitability but low investment may face transient earnings or limited growth options, which warrants higher required returns as compensation for that risk.

By outperforming the preexisting canonical factor models and reducing the number of unexplained statistically significant anomalies Hou *et al.* [1] is more consistent with the EMH by suggesting that these seemingly systematic mispricings are largely the product of an incomplete risk model rather than genuine market inefficiency.

However, because of the joint-hypothesis problem, this does not conclusively establish that the remaining anomalous returns are due to the incompleteness of the pricing model rather than the existence of market inefficiency.

A behavioural finance interpretation also seems viable. The negative relation between investment and subsequent returns could reflect investor overoptimism toward firms that invest aggressively, causing such firms to become overpriced and subsequently each low returns. Similarly, the profitability premium could arise if markets underreact to information about firms' fundamentals [6]. Under these views, the q-factor model may be absorbing behavioral anomalies into risk factors without truly identifying a risk-based mechanism.

To conclude, the q-factor model represents a significant methodological advance, it is theoretically grounded, empirically parsimonious and reduces unexplained cross-sectional return variation more effectively. Its consistency with the EMH is genuine given the validity of the underlying risk framework, a caveat that applies equally to all reduced-form factor models in asset pricing.

3.2 Comparison between HXZ 2015 and Fama-French 2018

Both papers use profitability to explain cross-sectional returns, but they differ in their choice of proxy. Fama & French [2] compares Operating Profitability (OP) and Cash Profitability (CP), arguing that these are preferable to return on equity because they sit lower on the income statement and therefore better isolate the underlying profitability of the firm. In particular, cash profitability removes the effect of accruals, so it is intended to strip out accounting noise and focus on more relevant components of profit. The issue in Fama & French [2] is not whether profitability matters, but which proxy produces the better profitability factor.

In Hou *et al.* [1], by contrast, the motivation for ROE is theoretical. Equation (4) in [1] implies that, holding investment fixed, firms with higher expected profitability should have higher expected returns. ROE is used as an observable proxy for this expected profitability, since it measures profits relative to the firm's equity base and is therefore closely tied to the model's economic mechanism. In this sense, Hou *et al.* [1] are not choosing ROE because it is simply the best empirical accounting ratio, but because it is the profitability measure that fits most naturally into their theoretical framework.

This difference in motivation helps explain the difference in construction. Hou *et al.* [1] use recent quarterly earnings information and update the ROE sorts monthly, because they want a profitability proxy that reflects the latest information about future profitability. This makes ROE particularly suited to capturing return patterns linked to earnings news, momentum, and related anomalies. Empirically, their ROE factor is strong; it earns an average of around 0.58% per month with a t-statistic of 4.81. They also show that it is strongly related to momentum, suggesting that momentum may partly proxy for profitability shocks. More broadly, the q-factor model performs well in digesting a large number of anomalies, so in Hou *et al.* [1] the profitability proxy matters because it is one of the central theoretically grounded drivers of expected returns.

In Fama & French [2], the discussion is more comparative. Their traditional profitability factor, RMW_o, is based on OP, while RMW_c is based on CP. The important result is that these two are not interchangeable: the correlation is only moderate (stated as $\rho = 0.66$), which suggests that accruals materially affect the operating profitability measure. Their main finding is that models using CP consistently outperform matching models using OP. Replacing RMW_o with RMW_c increase the model's maximum squared Sharpe ratio substantially (Out-of-sample avg. 0.108 vs 0.159 under the standard Fama-French six-factor model (FF5 + UMD)). The interpretation is that accruals add noise to operating profitability, whereas cash profitability better isolates the components of firm performance that is actually relevant for average returns.

The key contrast, then, is that Hou *et al.* [1] and Fama & French [2] are trying to capture different aspect of profitability. Hou *et al.* [1] want a proxy for expected profitability that fits the logic of q-investment theory, so ROE is attractive because it links naturally to investment, discount rates, and expected returns. Fama & French [2], however, are asking which accounting-based profitability proxy gives the strongest factor in practice, and their results favor CP over OP because it reduces accrual-related noise and increase maximum Sharpe ratio. Thus the disagreement is not simply about measurement error: it reflects two different priorities. ROE is chosen for theoretical consistency with investment-based pricing, while CP is chosen for empirical cleanliness and factor performance.

Overall, both papers agree that profitability is a fundamental dimension of expected returns, but they operationalise it differently. Hou *et al.* [1] favor ROE because it is a theoretically motivated proxy for expected profitability, where Fama & French [2] conclude that cash profitability is the superior empirical factor because it filters out accounting distortions more effectively than operating profitability.

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Appendix

A.1 Fama-MacBeth Process

Comparison between Original and Simplified Fama Macbeth Process

A.1.1 The Algebraic Equivalence

The key insight is that the Simplified Fama-MacBeth (FM) and Original FM share an identical **Pass 1**, and their Pass 2 estimators are algebraically equivalent in expectation.

Regression 1 is identical by construction. Both methods run the same time-series OLS for each portfolio i:

$$R_{it}^e = \alpha_i + \beta_{im} R_{mt}^e + \varepsilon_{it}$$

yielding the same $\hat{\beta}_{im}$ for every portfolio.

Regression 2 point estimates converge to the same quantity. The Simplified Fama-MacBeth regresses time-averaged returns on $\hat{\beta}_{im}$:

$$\bar{R}_i^e = \hat{\lambda}\hat{\beta}_{im} + \hat{a}_i$$

The Original Fama-MacBeth runs this regression at each t separately and averages, $\hat{\lambda}_{FM} = \frac{1}{T}\sum_{t=1}^{T}\hat{\lambda}_t$.By the Frisch-Waugh theorem and linearity of OLS, averaging T cross-sectional regressions over time produces the same coefficient as regressing the time-averaged dependent variable once.

$$\frac{1}{T}\sum_{t=1}^{T}\hat{\lambda}_t = \left(\hat{\beta}_m'\hat{\beta}_m\right)^{-1}\hat{\beta}_m' \qquad \frac{1}{T}\sum_{t=1}^{T}R_{it}^e = \left(\hat{\beta}_m'\hat{\beta}_m\right)^{-1}\hat{\beta}_m'\bar{R}^e$$

which is exactly the OLS estimator in the Simplified Fama-MacBeth. Hence $\hat{\lambda}_{\text{Simp}} = \hat{\lambda}_{\text{FM}}$ in finite samples, not just asymptotically.

A.1.2 Standard Errors

The only meaningful difference between the two methods, and it matters for inference is the difference between the Standard Errors. Simplified Fama-MacBeth uses standard OLS standard errors from the single cross-sectional regression as,

$$\widehat{\text{Var}}_{\text{Simp}}(\hat{\lambda}) = \hat{\sigma}^2\left(\hat{\beta}_m'\hat{\beta}_m\right)^{-1}$$

where $\hat{\sigma}^2$ is the cross-sectional residual variance. This treats the N portfolio observations as if they were i.i.d., which they are not portfolio returns are cross-sectionally correlated. Original Fama-MacBeth derives variance from the time-series dispersion of $\hat{\lambda}_t$:

$$\widehat{\text{Var}}_{\text{FM}}(\hat{\lambda}_{\text{FM}}) = \frac{1}{T^2}\sum_{t=1}^{T}\left(\hat{\lambda}_t - \hat{\lambda}_{\text{FM}}\right)^2$$

This is robust to cross-sectional correlation in ε_{it} because it never models the cross-sectional covariance structure directly, it lets the time-series variation of $\hat{\lambda}_t$ absorb it implicitly. The

resulting t-statistic is

$$t = \frac{\hat{\lambda}_{\text{FM}}}{\sqrt{\widehat{\text{Var}}(\hat{\lambda}_{\text{FM}})}} \sim N(0,1)$$

is asymptotically valid under much weaker assumptions than the Simplified Fama-MacBeth's OLS t-stat.

A.1.3 Consistency as $T \to \infty$

As shown in the cross-sectional framework, as $T \to \infty$:

$$\bar{R}_i^e \xrightarrow{p} E[R_{it}^e], \qquad \hat{\beta}_{im} \xrightarrow{p} \beta_{im}$$

so both estimators are consistent for the true risk premium $E[R_{mt}^e]$. The Simplified Fama-MacBeth gains nothing in terms of point estimate accuracy over the Original Fama-MacBeth and the convergence rate is the same. The difference is purely in **how reliably each method quantifies uncertainty around that estimate**.

A.1.4 Summary

In summary, both the Simplified FM and Original FM approaches produce identical first-stage estimates $\hat{\beta}_{im}$ by construction, as they rely on the same time-series regressions. Their second-stage estimates of the risk premium $\hat{\lambda}$ are also numerically equivalent.

Figure A.1: *Comparision Between Simplified and Original Fama-MacBeth Process*

The key distinction lies in the estimation of variance. As a result, only the Original Fama-MacBeth is robust to cross-sectional correlation in returns. Both methods remain consistent as $T \to \infty$, but inference differs in finite samples, where the Simplified FM fails to provide valid standard errors, while the Original FM delivers asymptotically valid inference. The practical implication is that Simplified FM will typically report smaller standard errors and larger t-statistics than Original FM, overstating the precision of $\hat{\lambda}$. In the figure A.1, we can observe the practical implications for some of the factors and all of the results matches with our theoretical analysis. We can also observe that, the risk premia estimates ($\hat{\gamma}$) are identical for both the cases and their difference is at machine-precision(1×10^{-16}) zero. Across the whole portfolio, the R^2 value is same for the methods due to the fact that, the data sample for our analysis so large which replicates the consistency.

Algorithm 1: Fama-MacBeth Cross-sectional Regression

Require: $\mathbf{F} \in \mathbb{R}^{T \times K}$ (factor returns), $\mathbf{R} \in \mathbb{R}^{T \times N}$ (portfolio returns)
Ensure: Risk premia estimates $\hat{\gamma}$, betas $\hat{\beta}$, gammas Γ

Regression 1: Time-Series

for each portfolio $n = 1, \ldots, N$ **do**
  Estimate OLS regression:

$$R_{n,t} = \alpha_n + \boldsymbol{\beta}_n^\top \mathbf{f}_t + \varepsilon_{n,t}, \quad t = 1, \ldots, T$$

  Store $\hat{\alpha}_n$, $\hat{\boldsymbol{\beta}}_n \in \mathbb{R}^K$, and R_n^2
end for
Collect $\hat{\mathbf{B}} = [\hat{\boldsymbol{\beta}}_1, \ldots, \hat{\boldsymbol{\beta}}_N]^\top \in \mathbb{R}^{N \times K}$

Regression 2: Cross-Sectional Regressions

for each time period $t = 1, \ldots, T$ **do**
  Collect portfolio returns $\mathbf{r}_t = (R_{1,t}, \ldots, R_{N,t})^\top \in \mathbb{R}^N$
  Estimate OLS regression (betas fixed from Regression 1):

$$\mathbf{r}_t = \gamma_{0,t}\mathbf{1} + \hat{\mathbf{B}}\,\boldsymbol{\gamma}_t + \boldsymbol{\eta}_t$$

  Store $\hat{\boldsymbol{\gamma}}_t = (\hat{\gamma}_{0,t}, \hat{\gamma}_{1,t}, \ldots, \hat{\gamma}_{K,t})^\top$
end for
Collect $\boldsymbol{\Gamma} = [\hat{\boldsymbol{\gamma}}_1, \ldots, \hat{\boldsymbol{\gamma}}_T]^\top \in \mathbb{R}^{T \times (K+1)}$

Risk Premia Estimation

Compute time-series mean:

$$\hat{\gamma}_k = \frac{1}{T} \sum_{t=1}^{T} \hat{\gamma}_{k,t}, \quad k = 0, 1, \ldots, K$$

Compute standard deviation:

$$\sigma_k = \sqrt{\frac{1}{T-1} \sum_{t=1}^{T} (\hat{\gamma}_{k,t} - \hat{\gamma}_k)^2}$$

Compute Fama-MacBeth standard error:

$$\widehat{\mathrm{SE}}(\hat{\gamma}_k) = \frac{\sigma_k}{\sqrt{T}}$$

Compute t-statistic:

$$t_k = \frac{\hat{\gamma}_k}{\widehat{\mathrm{SE}}(\hat{\gamma}_k)}$$

return $\{\hat{\gamma}_k,\ \sigma_k,\ \widehat{\mathrm{SE}}(\hat{\gamma}_k),\ t_k\}_{k=0}^{K},\ \hat{\mathbf{B}},\ \mathbf{\Gamma},\ \{R_n^2\}_{n=1}^{N}$

Algorithm 2: Simplified Fama-Macbeth Cross-Sectional Regression

Require: $\mathbf{F} \in \mathbb{R}^{T\times K}$ (factor returns), $\mathbf{R} \in \mathbb{R}^{T\times N}$ (portfolio returns)
Ensure: Risk premia $\hat{\boldsymbol{\gamma}}$, betas $\hat{\mathbf{B}}$, alphas $\hat{\boldsymbol{\alpha}}$

Regression 1: Time-Series

for each portfolio $n = 1, \ldots, N$ **do**

Estimate OLS regression:

$$R_{n,t} = \alpha_n + \boldsymbol{\beta}_n^\top \mathbf{f}_t + \varepsilon_{n,t}, \quad t = 1, \ldots, T$$

Store $\hat{\alpha}_n$, $\hat{\boldsymbol{\beta}}_n \in \mathbb{R}^K$, and R_n^2

end for

Collect:

$$\hat{\boldsymbol{\alpha}} = (\hat{\alpha}_1, \ldots, \hat{\alpha}_N)^\top \in \mathbb{R}^N, \qquad \hat{\mathbf{B}} = [\hat{\boldsymbol{\beta}}_1, \ldots, \hat{\boldsymbol{\beta}}_N]^\top \in \mathbb{R}^{N\times K}$$

Regression 2: Single Cross-Sectional

Compute time-averaged portfolio returns:

$$\bar{R}_n = \frac{1}{T}\sum_{t=1}^{T} R_{n,t}, \quad n = 1, \ldots, N$$

Collect $\bar{\mathbf{r}} = (\bar{R}_1, \ldots, \bar{R}_N)^\top \in \mathbb{R}^N$

Estimate single OLS cross-sectional regression (betas fixed from Regression 1):

$$\bar{\mathbf{r}} = \gamma_0 \mathbf{1} + \hat{\mathbf{B}}\,\boldsymbol{\gamma} + \boldsymbol{\eta}$$

Obtain estimates via OLS:

$$\begin{pmatrix} \hat{\gamma}_0 \\ \hat{\boldsymbol{\gamma}} \end{pmatrix} = \left(\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}}\right)^{-1} \tilde{\mathbf{B}}^\top \bar{\mathbf{r}}, \qquad \tilde{\mathbf{B}} = \left[\mathbf{1} \;\middle|\; \hat{\mathbf{B}}\right] \in \mathbb{R}^{N\times(K+1)}$$

Statistical Inference

Compute OLS standard errors:

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\gamma}}) = \hat{\sigma}_\eta^2 \left(\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}}\right)^{-1}, \qquad \hat{\sigma}_\eta^2 = \frac{\hat{\boldsymbol{\eta}}^\top \hat{\boldsymbol{\eta}}}{N - K - 1}$$

Compute t-statistics:

$$t_k = \frac{\hat{\gamma}_k}{\widehat{\mathrm{SE}}(\hat{\gamma}_k)}, \quad k = 0, 1, \ldots, K$$

return $\{\hat{\gamma}_k,\ \widehat{\mathrm{SE}}(\hat{\gamma}_k),\ t_k\}_{k=0}^{K},\ \hat{\mathbf{B}},\ \hat{\boldsymbol{\alpha}},\ \{R_n^2\}_{n=1}^{N}$

A.2 Statistical Data Validation

We performed Variance Inflation Factor (VIF) test on the independent factors of all the models to test for multicollinearity. This test is just to check for consistency in the data as the Fama-MacBeth procedure addresses cross-sectional inference differently from OLS, the risk premia estimates come from averaging many cross-sectional regressions, and the standard errors are computed from the time-series variation of those estimates, not from a single OLS fit where VIF tests are designed to apply.

Table A.1: *Variance Inflation Factors (VIF) by Model*

Model	**Feature**	**VIF**
Fama–French	Mkt-RF	1.1471
	SMB	1.1437
	HML	1.0033
Macroeconomic Factors	Div. Growth	1.0072
	DEF = LT Corp. − LT Govt.	1.2660
	TERM = ST Govt. − LT Govt.	1.2701
In-Sample PCA	In-Sample PC1	1.0006
	In-Sample PC2	1.0005
	In-Sample PC3	1.0003
Out-of-Sample PCA	Out-of-Sample PC1	1.1477
	Out-of-Sample PC2	1.0629
	Out-of-Sample PC3	1.1533

PCA models (in-sample and out-of-sample): PCA factors are constructed to be orthogonal to each other by design. The principal components have zero correlation, so multicollinearity is structurally impossible. VIF would be 1.0 for all factors trivially in theoretical and we can also observe the same from our analysis in the table A.1

Fama-French (MKT, SMB, HML): These are not orthogonal, and in principle you could run a VIF. However, the empirical correlation between Fama-French factors is well-documented to be low enough that multicollinearity is not a serious concern in standard samples.

Macroeconomic factors: These are the most plausible factors for a VIF check, since macro variables can be correlated. As we can see from the table A.1, the macroeconomic factors have a VIF close one meaning they are not correlated.

A.3 Fama-French 2018 data

Table A.2: *Comparison of six-factor models that include an operating profitability (OP) or a cash profitability (CP) factor: July 1963–June 2016*

Levels of $Sh^2(f)$:

		Full-sample		In-sample		Out-of-sample	
Model	Actual	Average	Median	Average	Median	Average	Median
Six-Factor Operating Profitability							
Mkt, SMB, HML, RMW_O, CMA, UMD	0.135	0.152	0.149	0.177	0.169	0.108	0.102
Mkt, SMB, HML_S, RMW_{OS}, CMA_S, UMD_S	0.199	0.217	0.213	0.244	0.236	0.169	0.162
Mkt, S-F, H-F, R_O-F, C-F, U-F	0.134	0.148	0.146	0.172	0.166	0.106	0.100
Mkt, S-F, H_S-F, R_{OS}-F, C_S-F, U_S-F	0.167	0.182	0.180	0.206	0.200	0.138	0.132
Mkt, S-F, L, W_O-F, A-F, D-F	0.111	0.127	0.124	0.152	0.144	0.085	0.079
Mkt, S-F, L_S-F, W_{OS}-F, A_S-F, D_S-F	0.174	0.192	0.189	0.220	0.212	0.144	0.137
Six-Factor Cash Profitability							
Mkt, SMB, HML, RMW_C, CMA, UMD	0.190	0.208	0.205	0.236	0.228	0.159	0.152
Mkt, SMB, HML_S, RMW_{CS}, CMA_S, UMD_S	0.226	0.244	0.241	0.274	0.265	0.194	0.186
Mkt, S-F, H-F, R_C-F, C-F, U-F	0.177	0.193	0.191	0.218	0.212	0.147	0.141
Mkt, S-F, H_S-F, R_{CS}-F, C_S-F, U_S-F	0.188	0.204	0.201	0.229	0.221	0.159	0.153
Mkt, S-F, L, W_C-F, A-F, D-F	0.160	0.176	0.173	0.204	0.196	0.128	0.121
Mkt, S-F, L_S-F, W_{CS}-F, A_S-F, D_S-F	0.210	0.229	0.226	0.259	0.250	0.177	0.170